

RAJAGOPALAN MADHYARJUNAN

Rolling Stock Engineering | Technical Validation | Program & Project Leadership
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PROFESSIONAL SUMMARY

A highly accomplished Rolling Stock Engineering Professional with over 23 years of progressive expertise in the engineering domain, including 13+ years of dedicated specialization in Metro, Mainline, and Suburban Electric Multiple Units (EMUs). Demonstrated success in steering the end-to-end product lifecycle—from bid technical evaluation and detailed product design to industrialization, change management, train-level verification, type testing, and service readiness. A strategic leader adept at managing high-performing Cross-Functional Teams (CFTs) to deliver complex engineering projects within strict Quality, Cost, and Delivery (QCD) benchmarks. Advanced proficiency in industry-leading CAD solutions (CATIA V5, Pro-E) and PLM/PDM ecosystems (Teamcenter, Windchill).

CORE COMPETENCIES & EXPERTISE

Area	Key Skills & Focus
Rolling Stock Lifecycle	Train Validation & Type Testing, Site Commissioning, Systems Engineering, Bid Technical Support, Requirements Management (DOORS), Integrated Service Readiness (ISR).
Mechanical Systems	Mechanical Fittings & Integration (MFI), Pneumatic Architecture Design, Underframe/Roof/Cab Integration, Vehicle Weight Management, GD&T, Pipework Integration.
Project & Process	Project Leadership, Change Management (CR), Six Sigma Green Belt, APQP Documentation (PFMEA, Control Plan, PFD), Kaizen & Poka-Yoke Implementation.
Technical Tools	CATIA V5, Pro-E Wildfire, SolidWorks, Teamcenter, Windchill, MS Office Suite (Excel, PowerPoint, Word).

PROFESSIONAL EXPERIENCE (ROLLING STOCK FOCUS)

- Alstom Transport | Lead Change Management – Monorail ProjectsApr 2026 – Present
- Change Governance:** Lead and execute complex Change Requests (CR) for global Monorail product variants, including prestigious projects in Sao Paulo, Cairo, Santiago, and Bangkok.
 - Impact Optimization:** Conduct rigorous technical and financial evaluations to estimate time, resource, and cost impacts across diverse design variants.
 - Field Engineering Resolution:** Formulate robust engineering solutions for active field issues, transforming field data into structured CRs and collaborating with CFTs to secure rapid closure.

Alstom Transport | ISR Project Lead (Agra & Kanpur Metro, Kanpur Depot) *Aug 2025 – Mar 2026*

- **Customer Handover Management:** Successfully spearhead the formal handover of critical project deliverables, including GA1 & GA2 spares, consumables, special tools, and test benches to the customer (GC/UPMRC).
- **Documentation & Compliance:** Manage assets for spares readiness while compiling, reviewing, and validating comprehensive supplier and technical documentation to ensure contract compliance.
- **Validation Oversight:** Direct the successful release of Milestone Completion Certificates (MCC) and lead technical demonstrations for subsystem overhauls to fulfill contractual obligations.

Alstom Transport | Product Introduction (PI) Lead (CMRL Project, Chennai) *May 2025 – July 2025*

- **Commissioning Excellence:** Executed the Miscellaneous Body Test (MBT) and engineered prompt solutions for complex electrical and mechanical anomalies to ensure flawless Product Introduction (PI) readiness at the depot.
- **Testing Coordination & Quality Assurance:** Synchronized testing schedules to secure vital external approvals, compiled comprehensive test reports, executed retrofits, and supervised monthly depot maintenance audits.

Alstom Transport | Domain Leader - Site Validation (Train Level Type Testing) *Sep 2022 – Apr 2025*

- **Strategic Validation Leadership:** Accountable for all Train-Level Validation and type testing methodologies executed at the Savli and Maneja sites across multiple rail platforms (Metro, Mainline EMU).
- **Team & Resource Optimization:** Managed an engineering team by defining key performance metrics, establishing clear annual objectives, and optimizing resource allocation against tight validation schedules.
- **Test Cycle Compression:** Spearheaded mechanical type tests while planning, structuring, and optimizing validation sequences to maximize efficiency and shorten total test cycle duration.
- **Work Package Accountability:** Author and approve Validation Management plans, technical compliance matrices, and evidence reports mapped against stringent customer requirements for TRAXX Locomotives.
- **Stakeholder Interface:** Serve as the primary technical point of contact for clients and external regulatory bodies, presenting validation findings, resolving complex technical queries, and securing final approvals.
- **Compliance Oversight:** Formally reviewed and authorized validation procedures and comprehensive final test reports to guarantee compliance for ICF-416 (Mainline EMUs) and diverse Metro platforms.

Alstom Transport (Formerly Bombardier Transportation) | Team Lead - Mechanical Fittings & Integration (MFI/PII) *July 2012 – Aug 2022*

- **Agra-Kanpur Metro MFI Team Lead:** Directed the end-to-end planning and technical execution of the MFI scope covering the cab, roof, underframe equipment, and critical pneumatic piping infrastructure.
- **Design Gate Milestones:** Guided the engineering team through rigorous phase-gate reviews, including Preliminary Design Reviews (PDR), Critical Design Reviews (CDR), and Detailed Design Reviews (DDR).
- **Production & Change Support:** Provided high-impact initial production support on the shop floor, rapidly mitigating risks and executing engineering Change Requests (CRs).
- **Aventra Platform (Derby, UK) Change Management Lead (Deputation):** Spearheaded the development, structuring, and implementation of a streamlined Change Request (CR) architecture to accelerate design improvements and resolve active manufacturing bottlenecks. Successfully eliminated CR backlogs, shortened development cycle times, and optimized industrialization throughput.

- **Movie Platform PII Engineer:** Pioneered the design of advanced pneumatic system architectures, introducing highly optimized, space-efficient structural layouts. Formulated inter-car system integration interfaces, managing space envelopes and precise mounting fixtures for all underframe systems.
- **Queensland EMU (QNGR) PII Engineer:** Owned the complete integration design for underframe equipment and associated structural interfaces. Conducted rigorous vehicle movement analyses and championed the critical vehicle weight management program to safeguard performance baselines.
- **Delhi Metro RS-11 PII Pneumatics Team Lead:** Engineered the optimization and redesign of pneumatic piping and exterior equipment integration during the critical early production phase. Championed engineering change management to resolve initial assembly bottlenecks and establish robust manufacturing readiness.
- **Bid Technical Support Lead:** Formulated highly technical bids for major multi-million dollar rolling stock prospects (including Kanchrapara SEMU/MEMU and BHIN Metro). Authored technical design proposals, detailed bid documentation, and innovative mounting methodologies while coordinating design reviews.
- **MRVC Phase 2 EMU:** Designed and developed the Shunting Desk, engineered ergonomics improvements for the Driver Desk, and optimized Electrical Control Cubicles and Cab Rear Wall Panels. Actively directed First Article Inspections (FAI) to validate production intent.

PREVIOUS ENGINEERING EXPERIENCE (FOUNDATION BUILDING)

Mahindra Engineering Services (Bangalore) | Senior Design Engineer – Construction Equipment

Jan 2009 – Jun 2012

- **Powertrain Integration:** Conceptualized and executed the packaging and installation design for Tier IV Volvo compactors and excavators.
- **Regulatory Compliance Upgrades:** Spearheaded the integration and upgrading of powertrain sub-systems to align with stringent international Tier 4 emissions compliance regulations.
- **Routing Design:** Designed and verified complex pneumatic/hydraulic piping routes and electrical wiring harness layouts using 3D CAD.

Infotech Enterprises Ltd. (Hyderabad) | Senior Design Engineer

Aug 2007 – Oct 2008

- **Product & Mechanical Design:** Managed the end-to-end design and development of electric toothbrush fixtures and performed advanced drivetrain analytics for high-volume consumer appliances.
- **Drivetrain & Eco Analysis:** Conducted detailed drivetrain kinematic studies and Eco-Scan sustainability assessments for household appliances.
- **Specialized Equipment Engineering:** Designed precision process machinery, including Halo Measurement microscopic devices, light-scanning exposure equipment, normalization programming machines, and product validation testers from initial concept to manufacture.

Pricol Ltd. (Coimbatore) | Senior Design Engineer – Automotive Components

Jan 2005 – Jun 2007

- **Product Development:** Engineered high-accuracy digital/analog speed sensors and dashboard instrument clusters for leading 2-wheeler and 4-wheeler vehicle OEMs.
- **Industrialization & Validation:** Coordinated the high-volume industrialization and testing validation protocols for commercialized products, including Piaggio instrument clusters.
- **Advanced Sensor Engineering:** Specialized in the design of high-sensitivity components, including Non-Contact Tacho Sensors, Yamaha Phonic wheels, and Maruti MRE sensors.

Bharat Technologies Auto Components Ltd. (Hosur) | Development Engineer Jan 2003 – Dec 2004

- **Prototype Manufacturing:** Directed the prototyping and manufacturing process of structural auto components (including two-wheeler chassis, mufflers, and exhaust systems) for flagship TVS Motor models (Victor GLX, Apache, Star).
- **Tooling & Jigs Design:** Conceptualized and engineered a diverse range of robotic welding jigs, production fixtures, Special Purpose Machines (SPMs), and inspection gauges.
- **Continuous Improvement (CI):** Validated tooling design through trial production runs, adapted existing SPMs to integrate engineering changes, and re-engineered shop-floor layouts to optimize flow.
- **Quality Assurance & APQP:** Prepared accurate capital expenditure costing for jigs/fixtures and authored full APQP lifecycles (including PFMEA, Control Plans, and Process Flow Diagrams).
- **Lean Manufacturing:** Actively deployed Pokay-Yoke (error-proofing) mechanisms and Kaizen continuous improvement frameworks.

Kali Material Handling Systems Ltd. (Kumbakonam) | Design Engineer Jun 2000 – Jan 2002

- **Bulk Material Handling Design:** Configured heavy-duty idlers, pulleys, and structural subsystems tailored for industrial belt conveyor networks.
- **Component Engineering:** Engineered specialized idler categories (Impact, Carrying, Return, Self-Aligning) alongside their respective heavy-gauge support frames.
- **Mechanical Calculations:** Executed structural design calculations, generated detailed process operation sheets, and approved fabrication drawings for the production floor.

KEY ACHIEVEMENTS & CONTRIBUTIONS

- **Corporate Recognition:** Honored with an engineering excellence award for resolving critical, highly complex MFI interface conflicts, preventing line stoppages, and safeguarding the industrialization timeline.
- **Process Cycle Time Reduction:** Standardized and optimized the Change Request (CR) workflow as Aventura Platform Lead, drastically reducing engineering backlogs and accelerating overall product development velocity.
- **Six Sigma Cost Optimization:** Capitalized on Six Sigma Green Belt methodologies to lead a high-visibility platform standardization project; maximized electrical component reuse and achieved a significant reduction in unique part numbers across active rail platforms.
- **Manufacturing Innovation:** Pioneered an advanced ultrasonic welding process design for speed sensor housings, establishing a highly repeatable manufacturing standard that yielded immediate, quantifiable cost reductions.

EDUCATION & PROFESSIONAL DEVELOPMENT

- **Diploma in Product Design** | MS Ramaiah School of Advanced Sciences, Bengaluru (2010)
- **MBA in Marketing** | Annamalai University, Chidambaram (2007)
- **B.E. in Mechanical Engineering** | AAM Engineering College, Bharathidasan University (1999)
- **Diploma in Mechanical Engineering** | Shanmuga Polytechnic, Thanjavur (1996)

PERSONAL DETAILS

- **Languages:** Tamil (Native), English (Fluent), Hindi (Proficient).
- **Interests:** Travel, Reading, Yoga, Cricket, Table Tennis.
- **Personal Status:** Married with two children.